

AKHudfC Excel Add-In UDF General Information

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This is a basic Excel add-in written with Csharp / ExcelDNA with the intent the assistance of the preparation of engineering calculations .

Installation

The appropriate add-in file (*.XLL) can be downloaded from GitHub.

<https://github.com/debear81/AKHudfC>

Once the file has been downloaded, place the file in the appropriate Excel Add-In folder and load it to Excel.

User Defined Function (UDF) Descriptions

This add-in contains 17 UDFs (not all of which are working properly yet).

All functions are prepended by "C_" for ease of location and input into cells.

W - Function is Working N - Function is NOT working E - Function is/was experimental

Index	Function	Description	W	N	E
1	C_AbsMax	Returns the absolute maximum (farthest from zero) of the entered numbers/references.	X		
2	C_AbsMin	Returns the absolute minimum (closest from zero) of the entered numbers/references.	X		
3	C_BiggerNumber	Function to track the milliseconds required to count up to a number.			X
4	C_ChangeCase	Changes the case (upper, lower, sentence) of a string.			X
5	C_Clamplt	Returns the specified value if it falls within the specified range, otherwise returns a limiting value.	X		
6	C_CountUnique	Counts the number of unique cell values in a range.	X		
7	C_FootToMix	Converts a number (assumed to be feet) to a mixed dimension string (ft-in).	X		
8	C_Formula	Similar to FORMULATEXT function, returns string of referenced cell's equation, showing values.	X		
9	C_InchToMix	Converts a number (assumed to be inch) to a mixed dimension string (ft-in).	X		
10	C_Indirect	Similar to INDIRECT function, but could access closed workbooks.		X	
11	C_IsBetween	Returns a true/false boolean, indicating if a value falls within a specified range.	X		
12	C_Linterp	Performs a linear interpolation, based on two known points (X, Y).	X		
13	C_MixToInch	Converts a mixed unit (ft-in) string to a decimal number of inches.	X		
14	C_Mmatch	Returns the matching row number of a table, based on multiple criteria.	X		
15	C_RevStr	Reverses the order of a specified string.	X		
16	C_SheetName	Returns the worksheet name of the active cell.	X		
17	C_WorkbookName	Returns the workbook name of the active cell.	X		

AKHudfC Excel Add-In UDF descriptions**C_AbsMax (ranges), C_AbsMin(ranges)**

Returns the value with the largest or smallest absolute magnitude from the supplied values.

C_AbsMax returns the value farthest from zero.

C_AbsMin returns the value closest to zero.

Inputs may be individual values, cell references, or ranges. Up to 9 arguments may be supplied.

Sample Use

argument	2.000	-2.500	101.000	-0.101
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C_AbsMax 101 = C_AbsMax(2.000,-2.500,101.000,-0.101)

C_AbsMax 101 = C_AbsMax(2.000:-0.101)

C_AbsMin -0.101 = C_AbsMin(2.000,-2.500,101.000,-0.101)

C_AbsMin -0.101 = C_AbsMin(2.000:-0.101)

Capabilities

Accepts numbers, cell references, and ranges.

Returns the original signed value, not the absolute value.

Example: if -10 is farthest from zero, the function returns -10, not 10.

Limitations

Blank cells / text values are ignored .

If two values tie, returns the first matching value found.

Does not preserve source-cell number formatting.

C_BiggerNumber(BigNumber, Step)

Experimental timing function used to compare approximate calculation speed between a compiled C# Excel-DNA UDF and a comparable VBA UDF.

This function counts from zero to BigNumber using Step as the increment, then returns the elapsed time as a text string.

VBA CODE:

```
Public Function v_BiggerNumber(BigNumber As Double, Optional StepSize As Double = 1) As String
    Dim counter As Double, Dim startTime As Single, Dim endTime As Single, Dim elapsedTime As Single

    counter = 0
    startTime = Timer ' Time in seconds (with decimal precision)
    Do While counter <= BigNumber
        counter = counter + StepSize
    Loop
    endTime = Timer
    elapsedTime = endTime - startTime
    If elapsedTime < 0 Then elapsedTime = elapsedTime + 86400
    v_BiggerNumber = "This took " & Format(elapsedTime * 1000, "0") & " milliseconds."
End Function
```

Sample Use

BigNumber = 900,000.0

Step = 0.25

C_BiggerNumber = This took 8 milliseconds. = **C_BiggerNumber(900,000.0,0.25)**

V_BiggerNumber = This took 63 milliseconds. = **v_BiggerNumber(900,000.0,0.25)**

Capabilities

Accepts a target number and step size.

Returns approximate elapsed processing time in milliseconds.

Intended for development/testing only.

Limitations

Not intended for production calculations.

Results vary by computer, Excel session, processor load, and recalculation state.

Step must be positive; zero or negative values may cause an infinite loop.

Very small step values or very large target values may cause long calculation times.

C_ChangeCase(String, Case)

Test string function that returns the supplied text with the requested capitalization style.

Case = 0 returns the original text.

Case = 1 converts text to uppercase.

Case = 2 converts text to lowercase.

Case = 3 converts text to sentence case.

Any other value defaults to 0.

Sample Use

String = **AaBbCcDdEe**

Case = **0 1 2 3 15**

C_ChangeCase = AaBbCcDdEe = C_ChangeCase(AaBbCcDdEe,0)

C_ChangeCase = AaBBCCDDEE = C_ChangeCase(AaBbCcDdEe,1)

C_ChangeCase = aabbccddeee = C_ChangeCase(AaBbCcDdEe,2)

C_ChangeCase = Aabbccddeee = C_ChangeCase(AaBbCcDdEe,3)

C_ChangeCase = AaBbCcDdEe = C_ChangeCase(AaBbCcDdEe,15)

Capabilities

Accepts text strings or cell references containing text.

Preserves the original text when Case = 0 or an unrecognized case value is entered.

Limitations

Sentence case is basic: likely only capitalizes the first character and lowercases the rest.

Does not apply title case.

Does not detect multiple sentences, abbreviations, or proper nouns.

C_ClampIt (Value, MinValue, MaxValue)

Returns Value if it falls within the specified minimum and maximum limits.

If Value is less than MinValue, the function returns MinValue.

If Value is greater than MaxValue, the function returns MaxValue.

Intended for engineering calculations where a computed value must be kept within specified minimum and maximum limits.

Sample Use

Value	5.00
MinValue	2.50
MaxValue	10.00

Value	4.50
MinValue	6.50
MaxValue	10.00

C_ClampIt = 5 = C_ClampIt(5.00,2.50,10.00)

6.5 = C_ClampIt(4.50,6.50,10.00)

Capabilities

Useful for limiting calculated values to a code-specified range.

Accepts numeric values or cell references.

Includes the limit values as acceptable values.

Limitations

MinValue should be less than or equal to MaxValue.

Non-numeric inputs return an error.

Does not indicate whether the original value was adjusted; it only returns the limited value.

C_CountUnique (Range)

Function to count the number of distinct values in a specified range of cells.

Sample Use

Range	5.00	5.00	1.00	1.00
	1	1.00	1.00	1.00
	1	1.00	5.00	1.00

C_CountUnique = 2 =C_CountUnique(C152:F154)

Range	BEAM	beam	beam	brace
	BEAM		beam	brace
	BEAM	beam	beam	brace

C_CountUnique = 4 =C_CountUnique(C158:F160)

Capabilities

Accepts a single cell range.

Counts each repeated value only once.

Works with numeric values and/or text values, depending on your actual function behavior.

Limitations

Blank cells are counted.

Text comparisons is case-sensitive.

Values that appear equal but are stored differently may be treated differently, such as 1, 1.0, and "1".

C_FootToMix(Foot)

String manipulation function that translates a decimal number of an assumed dimension (ft) into a string of mixed dimensions (ft-in).

Foot = decimal feet input.

FrcRnd = inch fraction rounding increment, such as 1/16, 1/8, or 0.125. Default is 1/2".

txtBool = output format option:

0 = symbolic format, e.g. 2'-7-1/4", 1 = text format, e.g. 2 ft 7-5/16 in, defaults to text format.

Sample Use

ft	inch	inch
2	7	0.3125

number = 2.609375 = 2+(7+0.3125)*1/12

FrcRnd = 0.125 1/16 (empty cell)

txtBool = 0 1 15

C_FootToMix = 2'-7-1/4" 2 ft 7-5/16 in 2 ft 7-1/2 in = C_FootToMix(2.609375,0 ,15)

Capabilities

Converts decimal feet to feet, inches, and fractional inches.

Allows user-specified rounding precision.

Provides symbolic or text-based output format.

Limitations

Returns text, not a numeric value.

Assumes the input value is in feet.

Rounding may slightly change the displayed dimension.

Clarify how negative values are handled, if supported.

C_Formula(Cell)

Similar to Excel's native FORMULATEXT function, but replaces referenced cells with their values. This can make calculation sheets easier to review by showing the substituted values used in the formula.

Sample Use

	a	b	c	d	e
	10	25.06544	0.65	3/4	yourtext
	<u>Excel Formulatext</u>		<u>C_Formula</u>		
C_Formula =	35.07	=C223+D223	10+25.06544		
C_Formula =	38.288	=D223/E223	25.06544/0.65		
C_Formula =	3.16227766	=SQRT(C223)	SQRT(10)		
C_Formula =	yourtext	=G223	yourtext		

Capabilities

Accepts a single cell reference.

Displays formulas with referenced cell values substituted in place of cell addresses.

Useful for calculation documentation and backchecking.

Limitations

Returns text only.

May not fully resolve complex formulas, nested functions, named ranges, external workbook references, or references to other sheets.

Displayed values may depend on cell formatting/rounding.

If the referenced cell does not contain a formula, the function returns the cell value as text.

C_InchToMix(Inch)

String manipulation function that translates a decimal number of assumed dimension (inch) into a string of mixed dimensions (ft-in).

Inch = decimal inches input.

FrcRnd = inch fraction rounding increment, such as 1/16, 1/8, or 0.125.

txtBool = output format option:

0 = symbolic format, e.g. 2'-7-1/4"

1 = text format, e.g. 2 ft 7-5/16 in

other/blank = default format is 1/2".

Sample Use

	ft	inch	inch	
		2	7	0.3125

number = 31.3125 = 2*12/1+(7+0.3125)

FrcRnd = 0.125 1/16 (empty cell)

txtBool = 0 1 15

C_InchToMix = 2'-7-1/4" 2 ft 7-5/16 in 2 ft 7-1/2 in = C_InchToMix(31.3125,0 ,15)

Capabilities

Converts decimal inches to feet, inches, and fractional inches.

Allows user-specified fractional rounding.

Provides symbolic or text-based output format.

Limitations

Returns text, not a numeric value.

Assumes input is in inches.

Rounding may slightly change the displayed dimension.

Negative-value behavior not supported.

C_Indirect()

NOT FUNCTIONING AS INTENDED YET.

Eventual intent is to operate similar to the INDIRECT.EX function of the MoreFunc add-in, which would interact with separate, closed Excel workbooks.

C_IsBetween (Value, MinValue, MaxValue)

Returns TRUE if Value falls within the specified minimum and maximum limits. Otherwise, returns FALSE.

Limit values are included. A value equal to MinValue or MaxValue returns TRUE.

Sample Use

Value	5.00
MinValue	2.50
MaxValue	10.00

Value	4.50
MinValue	6.50
MaxValue	10.00

C_IsBetween = TRUE = C_IsBetween(5.00,2.50,10.00) FALSE = C_IsBetween(4.50,6.50,10.00)

Capabilities

Accepts numeric values or cell references.

Useful for checking whether a calculated value is within an allowable range.

Can be combined with IF, AND, or other Excel functions.

Limitations

MinValue should be less than or equal to MaxValue.

Non-numeric inputs return an error.

Returns only TRUE/FALSE; it does not adjust the value. Use C_ClampIt if you want the value limited to the range.

C_Linterp(x, x1, y1, x2, y2)

Performs linear interpolation to estimate the y value corresponding to a specified x value between two known points.

x = x-value where the interpolated result is desired.

x1, y1 = first known point.

x2, y2 = second known point.

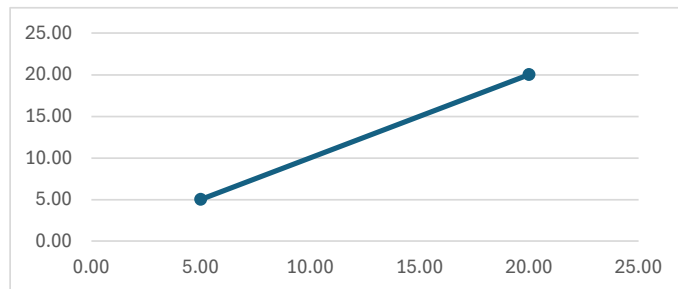
Sample Use

x = 12.50

x = 25.00

y = desired value

	<u>Known x</u>	<u>Known y</u>
1	5.00	5.00
2	20.00	20.00



C_Linterp = 12.50 = C_Linterp(12.50,5.00,5.00,20.00,20.00)

C_Linterp = 25.00 = C_Linterp(25.00,5.00,5.00,20.00,20.00)

Capabilities

Interpolates between two known (x, y) points.

Can also extrapolate if x is outside the known range, if your function allows that.

Useful for table lookups, code values, and engineering charts.

Limitations

x1 and x2 cannot be equal.

Assumes a straight-line relationship between the two known points.

Does not check whether interpolation is appropriate for the underlying data.

If extrapolation is allowed, results outside the known range should be used with caution.

C_MixToInch(String)

Converts a mixed feet-inch dimension string to a decimal inch value.

This function is intended to recognize common feet-inch formats such as 3'-7 1/4" or 3'-7-1/4" and return the equivalent decimal inches.

Sample Use

number = 3' 7 1/4" 3'-7 1/4" 3'-7-1/4" 3ft 7 1/4in 3ft-7-1/4in 3 feet 7 1/4 inches

C_MixToInch = 43.250 43.250 43.250 43.250 43.250 43.250

Capabilities

Converts mixed feet-inch text to decimal inches.

Intended for common structural/architectural dimension formats.

Output is numeric and can be used in other calculations.

Limitations

Experimental / not fully reliable yet.

Input formatting is currently sensitive.

Some text formats may not parse correctly.

Verify results before using in final calculations.

Negative dimensions, decimal inches, or unusual spacing may not be supported yet.

C_Mmatch(MatchRange1, Criteria 1, [MatchRange2], [Criteria2], [MatchRange3], [Criteria 3], [MatchRange4], [Criteria4])

Returns the row number of the first table row that matches the specified criteria. This is similar to MATCH, but allows multiple criteria to be checked at the same time.

Sample Use

criteria1 **103** (Member ID)

criteria2 **LC2** (Load Case)

c_mmatch = 10 =C_MMatch(\$C\$401:\$C\$414,C395,\$E\$401:\$E\$414,C396)

Row No.	Member ID	Mem Type	Load Case	Moment (k-ft)	Shear (k)
1	101	Beam	LC1	10.5	3.5
2	102	Beam	LC1	12.7	7.5
3	103	Beam	LC1	16.4	4.3
4	104	Beam	LC1	9.6	8.9
5	105	Beam	LC1	8.5	4.5
6	106	Brace	LC1	3.5	6.7
7	107	Brace	LC1	6.3	9.7
8	101	Beam	LC2	10.1	3.1
9	102	Beam	LC2	12.3	7.1
10	103	Beam	LC2	16	3.9
11	104	Beam	LC2	9.2	8.5
12	105	Beam	LC2	8.1	4.1
13	106	Brace	LC2	3.1	6.3
14	107	Brace	LC2	5.9	9.3

Capabilities

Supports up to 4 match criteria.

Accepts one-dimensional match ranges.

Useful for finding a row in engineering tables based on member number, load case, material type, etc.

Returns the matching row number for use in other lookup formulas.

Limitations

Match ranges must be the same size and orientation.

Returns the first matching row if multiple rows match.

Does not return the matched value itself; use the returned row number with another lookup/index function.

Blank cells, text/number mismatches, and case sensitivity should be verified.

C_RevStr(String)

Reverses the character order of the supplied text string.

Sample Use

String = CAT

	<u>Excel Formul</u> <u>atext</u>	<u>C Formula</u>
C_RevStr = TAC	=C_RevStr(C437)	C_RevStr(CAT)
C_RevStr = FART	=C_RevStr("TRAF")	C_RevStr("TRAF")

Capabilities

Accepts a text string or cell reference containing text.
Returns the reversed string as text.
Useful for testing string handling or helper routines.

Limitations

Returns text only.
Does not preserve formatting from the source cell.
Spaces and punctuation are reversed along with all other characters

C_SheetName(), C_WorkbookName()

Returns the worksheet name or workbook name for the cell where the function is entered.

Sample Use

	<u>Formulat</u> <u>ext</u>	<u>C Formula</u>
C_SheetName = UDFs	=C_SheetName()	C_SheetName()
C_WorkbookName = AKHudfC addin UDFs.xlsx	=C_WorkbookName()	C_WorkbookName()

Capabilities

Requires no input arguments.
C_SheetName() returns the current worksheet name.
C_WorkbookName() returns the current workbook file name.
Useful for calculation headers, print titles, and documentation sheets.

Limitations

Returns text only.
Result may not update immediately after renaming a sheet or workbook unless the worksheet recalculates.
Workbook name may be different before the file has been saved.